PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE B · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

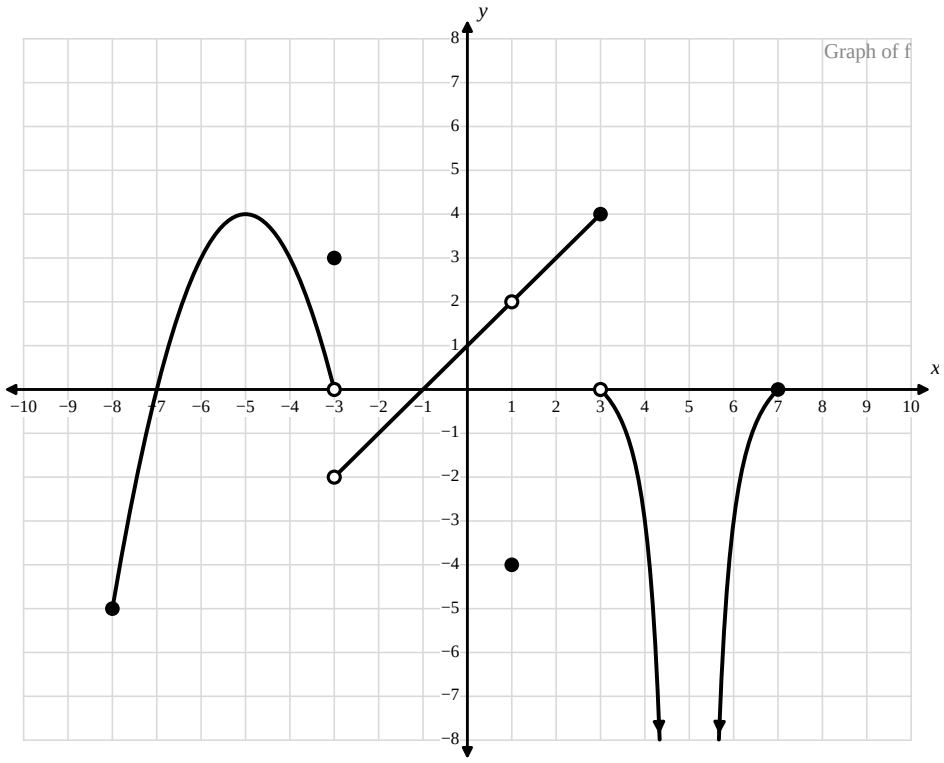
_____

_____

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

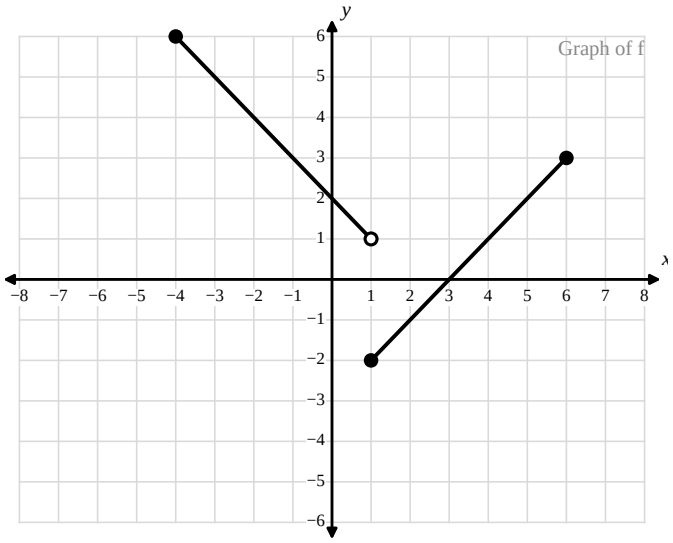


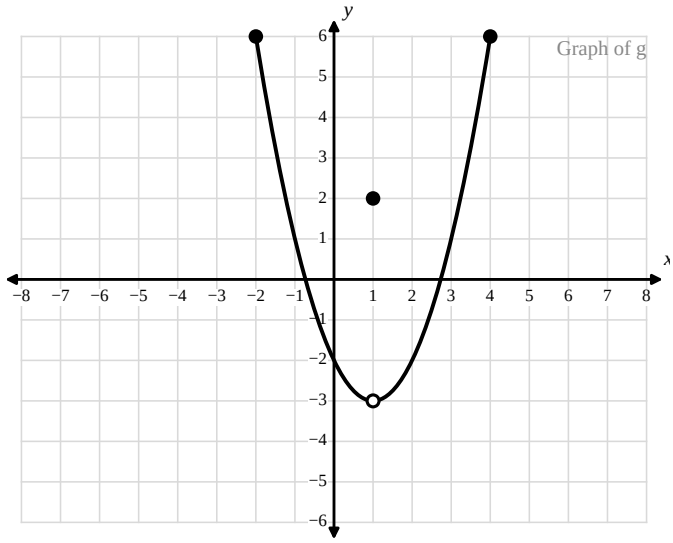
1. $\lim_{x\to 1} f(x) =$	**2.** $f(1) =$	**3.** $\lim_{x\to -3} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 0$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 1} [f(x) + g(x)]$

(2 points)

6. Use the table to estimate $\lim_{x \to -1} f(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−1.1	−1.01	−1.001	$\to -1 \leftarrow$	−0.999	−0.99	−0.9
$f(x)$	3.21	3.0201	3.002001		−0.998	−0.98	−0.8

$\lim_{x \to -1^-} f(x) \approx$ _____ $\lim_{x \to -1^+} f(x) \approx$ _____ $\lim_{x \to -1} f(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x \to 3} \dfrac{x+2}{x-3}$

8. $\lim\limits_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$

9. $\lim\limits_{x \to 6} \dfrac{\sqrt{x+3} - 3}{x - 6}$

10. $\lim\limits_{x \to -1} \dfrac{\frac{1}{x+3} - \frac{1}{2}}{x+1}$

11. For $f(x) = \sqrt{x-2}$, find $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

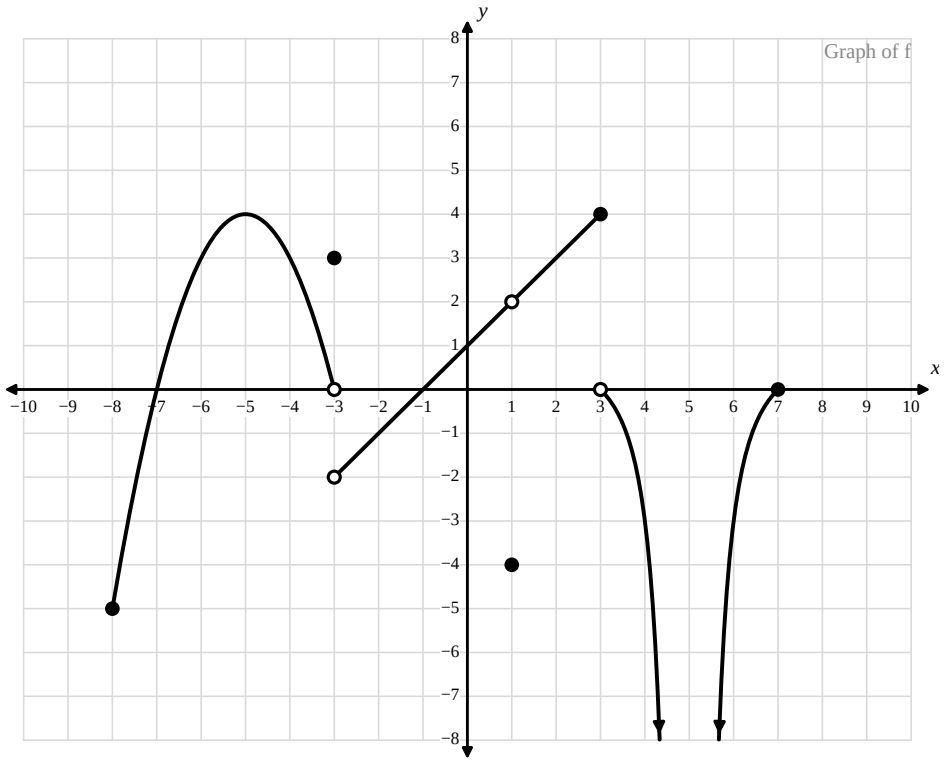
PRACTICE B · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to 1} f(x) =$

Both sides of the line head to the open circle at $(1, 2)$.

2

2. $f(1) =$

The filled dot at $x = 1$ *is at height* -4.

-4

3. $\lim_{x\to -3} f(x) =$

$\lim_{x\to -3^-} f(x) = 0$,
$\lim_{x\to -3^+} f(x) = -2$
$0 \neq -2$, *so* $\lim_{x\to -3} f(x)$ *does not exist.*

DNE

(2 points)

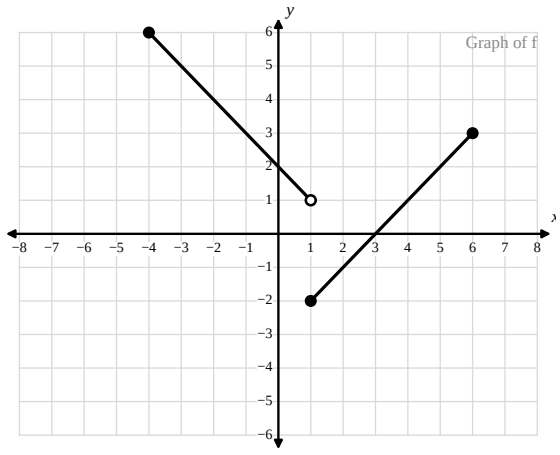
4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 0$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

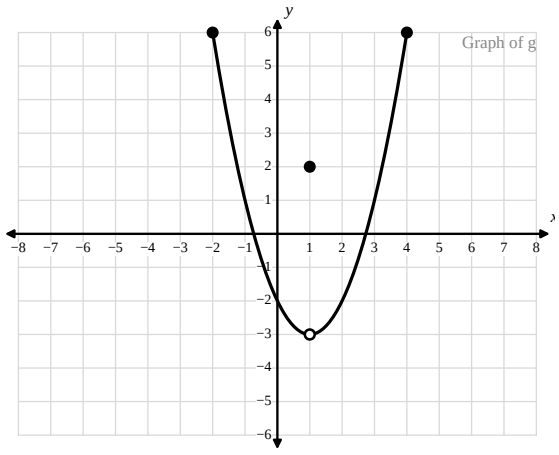
Height 0 is reached from both sides at $x = -7$ *and* $x = -1$.

Not $x = -3$ *(left side only), not* $x = 3$ *(right side only; the left heads to 4), not* $x = 7$ *(graph ends there).*

$a = -7,\ a = -1$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 1} [f(x) + g(x)]$

$\lim_{x \to 1^-} [f(x) + g(x)] = 1 + (-3) = -2$

$\lim_{x \to 1^+} [f(x) + g(x)] = -2 + (-3) = -5$

$-2 \neq -5$, *so* $\lim_{x \to 1} [f(x) + g(x)]$ *does not exist.*

DNE

(2 points)

6. Use the table to estimate $\lim_{x \to -1} f(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−1.1	−1.01	−1.001	→ −1 ←	−0.999	−0.99	−0.9
$f(x)$	3.21	3.0201	3.002001		−0.998	−0.98	−0.8

Left: 3.21, 3.0201, 3.002001

Right: $-0.8,\ -0.98,\ -0.998$

The two sides head to different values, so the table suggests the limit does not exist.

DNE (suggested)

$\lim_{x \to -1^-} f(x) \approx$ 3 $\lim_{x \to -1^+} f(x) \approx$ −1 $\lim_{x \to -1} f(x) \approx$ DNE

Evaluate each limit. (2 points each)

7. $\lim_{x\to 3} \dfrac{x+2}{x-3}$

Substitution produces $\frac{5}{0}$: a nonzero number over 0.

Left: top near 5, bottom tiny negative, so $-\infty$. Right: $+\infty$.

DNE

8. $\lim_{x\to 2} \dfrac{x^3-8}{x^2-4}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 2} \frac{x^3-8}{x^2-4} = \lim_{x\to 2} \frac{(x-2)\left(x^2+2x+4\right)}{(x-2)(x+2)}$$
$$= \lim_{x\to 2} \frac{x^2+2x+4}{x+2}$$
$$= \frac{4+4+4}{2+2} = \frac{12}{4} = 3$$

3

9. $\lim_{x\to 6} \dfrac{\sqrt{x+3}-3}{x-6}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 6} \frac{\sqrt{x+3}-3}{x-6} = \lim_{x\to 6} \frac{(x+3)-9}{(x-6)\left(\sqrt{x+3}+3\right)}$$
$$= \lim_{x\to 6} \frac{x-6}{(x-6)\left(\sqrt{x+3}+3\right)} = \lim_{x\to 6} \frac{1}{\sqrt{x+3}+3}$$
$$= \frac{1}{\sqrt{9}+3} = \frac{1}{6}$$

$\frac{1}{6}$

10. $\lim_{x\to -1} \dfrac{\frac{1}{x+3}-\frac{1}{2}}{x+1}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to -1} \frac{\frac{1}{x+3}-\frac{1}{2}}{x+1} = \lim_{x\to -1} \frac{2-(x+3)}{2(x+3)(x+1)}$$
$$= \lim_{x\to -1} \frac{-(x+1)}{2(x+3)(x+1)}$$
$$= \lim_{x\to -1} \frac{-1}{2(x+3)} = \frac{-1}{2(2)} = -\frac{1}{4}$$

$-\frac{1}{4}$

11. For $f(x) = \sqrt{x-2}$, find $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$

$$\lim_{h\to 0} \frac{\sqrt{x+h-2}-\sqrt{x-2}}{h}$$
$$= \lim_{h\to 0} \frac{(x+h-2)-(x-2)}{h\left(\sqrt{x+h-2}+\sqrt{x-2}\right)}$$
$$= \lim_{h\to 0} \frac{h}{h\left(\sqrt{x+h-2}+\sqrt{x-2}\right)} = \lim_{h\to 0} \frac{1}{\sqrt{x+h-2}+\sqrt{x-2}}$$
$$= \frac{1}{2\sqrt{x-2}}$$

$\frac{1}{2\sqrt{x-2}}$